

The constant $A(p)$ in subcritical Galton–Watson branching

and the Koenigs obstruction to an elementary closed form

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Abstract

For the binary Galton–Watson process with division probability $p \in [0, \frac{1}{2})$, the survival probability S_n satisfies $S_n \sim A(p) (2p)^n$, so the limiting mean population size conditional on non-extinction is $1/A(p)$. We develop the basic branching background, the definition and properties of $A(p)$, an infinite-product representation, near-critical asymptotics, and numerical/search attempts at a closed form. We then identify $A(p)$ with (twice) the Koenigs function of the logistic map $f_r(w) = rw(1 - w)$ at $w = \frac{1}{2}$ and multiplier $r = 2p$. Drawing on classical linearisation theory and the hypertranscendence results of Becker and Bergweiler, we explain why no elementary closed form for the conjugacy—and hence none expected for $A(p)$ —exists for $r \in (0, 1)$, while recording the exceptional elementary cases $r = 2$ and $r = 4$ (outside the branching range) in an appendix. The claim is stated carefully: fixed-multiplier non-elementarity of the conjugacy underpins the obstruction; a full classification of the scalar map $p \mapsto A(p)$ is not claimed.

Contents

1	Introduction	3
1.1	Certainty of extinction and expected lifetime for critical GW branching	5
1.1.1	Certainty of extinction	5
1.1.2	Expected lifetime	7
2	Limiting Conditional Mean $\mathbb{E}(Z_n Z_n > 0) = 1/A(p)$	9
3	Attempting to compute $A(p)$	10
3.1	Derivation of $A(p)$ as an infinite product	11
3.1.1	Convergence of the product	12
3.1.2	Advantages of the infinite product formula.	12
3.2	Asymptotic formula for $A(p)$ as $p \rightarrow \frac{1}{2}^-$	12
3.3	Genetic algorithm search attempts	13
4	Relationship to the Koenigs function	15
4.1	From the survival recurrence to the logistic map	15
4.2	Koenigs linearisation	16
4.3	The identity $A(p) = 2 \lim r^{-n} f_r^{cn}(\frac{1}{2})$	16
5	No elementary closed form for the conjugacy when $r \in (0, 1)$	17
5.1	Statement	17
5.2	Consequences for $A(p)$	18

6	Practical computation of $A(p)$ and $1/A(p)$	18
7	Conclusion	19
A	Elementary Koenigs functions at $r = 2$ and $r = 4$	20
A.1	Case $r = 2$ (logarithmic)	20
A.2	Case $r = 4$ (inverse trigonometric)	20
A.3	Affine conjugacy to $z^2 + c$	20
A.4	Becker–Bergweiler dichotomy (summary)	20

1 Introduction

The Galton–Watson process is a discrete-time, discrete-space Markov chain. It is a stochastic model in which every member of a population independently either replicates or dies at each generational increment; by these rules, the population fluctuates over the course of its lifetime. Originally formulated by Francis Galton and Henry Watson in the late 19th century, it stands as one of our most fundamental models of stochastic branching.

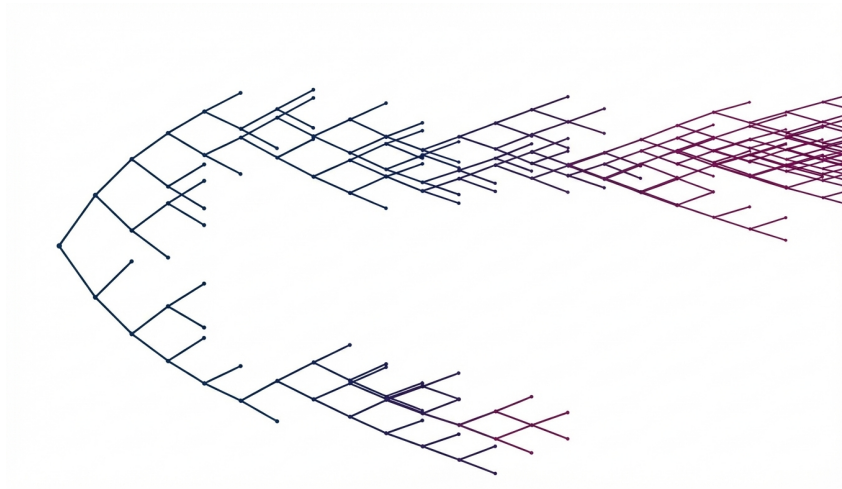
Let Z_n denote the population census at generation n . In the standard case, there is a single initial individual,

$$Z_0 = 1.$$

At the passage to each successive generation n , all Z_{n-1} existing individuals independently

- **replicate** with probability p , or
- **die** with the remaining likelihood $1 - p$.

This is the *simple* Galton Watson process. It is also possible to define the process as having potentially multiple offspring per individual, but here we focus on the classic binary case.



Realisation of the simple Galton-Watson process with $p = 0.58$.

Generally, the expected value of a discrete random variable X is defined by

$$\mathbb{E}(X) = \sum_k \mathbb{P}(X = x_k) x_k,$$

the sum over all possible states x_k in product with their probability of realisation.

Hence, after one generation (with $Z_0 = 1$), we have

$$\mathbb{E}(Z_1) = \mathbb{P}(Z_1 = 2) \cdot 2 + \mathbb{P}(Z_1 = 0) \cdot 0.$$

Replication occurs with probability p , resulting in two offspring. So this simplifies to

$$\mathbb{E}(Z_1) = 2p.$$

By the following argument, we can compute the expected population size at generation n : $\mathbb{E}(Z_n)$.

If we know the value of Z_n , then the conditional expectation of the next generation is

$$\mathbb{E}(Z_{n+1} \mid Z_n) = \sum_{i=1}^{Z_n} \mathbb{E}(Z_1) = 2p Z_n.$$

More generally, if we do not know Z_n , the tower property of conditional expectation gives

$$\mathbb{E}(Z_{n+1}) = \mathbb{E}(\mathbb{E}(Z_{n+1} \mid Z_n)) = \mathbb{E}(2p Z_n) = 2p \mathbb{E}(Z_n).$$

Hence,

$$\mathbb{E}(Z_2) = 2p \mathbb{E}(Z_1) = (2p)^2,$$

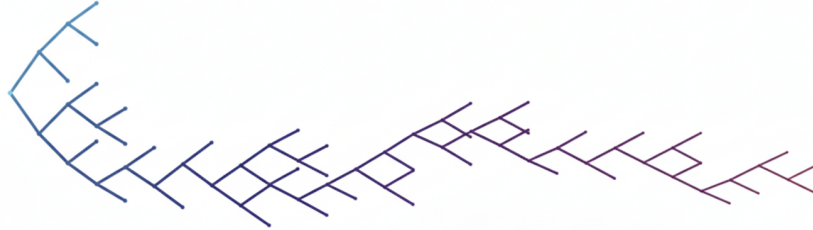
$$\mathbb{E}(Z_3) = 2p \mathbb{E}(Z_2) = (2p)^3,$$

so, induction gives

$$\mathbb{E}(Z_n) = (2p)^n.$$

From this result, we observe a standard behaviour of branching processes:

- If $2p > 1$, then $\mathbb{E}(Z_n)$ grows geometrically with n . In this case, the branching process is said to be *supercritical*.
- If $2p < 1$, then $\mathbb{E}(Z_n)$ decays geometrically with n , and the branching process is *subcritical*.



Subcritical Galton-Watson realisation with $p = 0.47$.

When $2p = 1$ (that is, $p = 0.5$), the Galton–Watson process is said to be *critical*. And in this case, the expectation remains constant:

$$\mathbb{E}(Z_n) = 1^n = 1 \quad \text{for all } n.$$

However, this does *not* mean that the population survives forever. Although the expectation remains positive for all finite generations, as in the supercritical case, when $2p = 1$, ultimate extinction is certain. So the process will eventually fixate at 0, but the expected lifetime of the process is infinite.

And this is a curious mathematical result...

Intuitively, it can be understood by the fact that the total lifetime, T , of this critical branching process is distributed according to a power law.

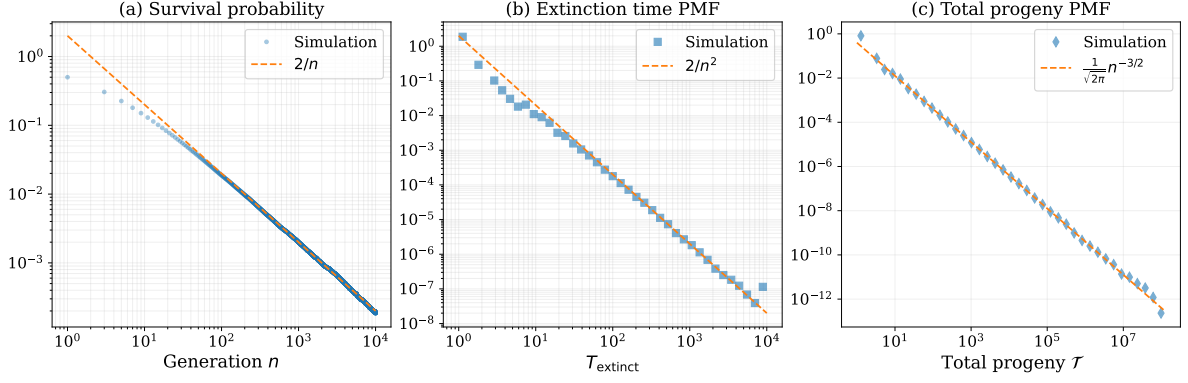


Figure 1: Power-law behavior in a critical Galton–Watson branching process ($p = 0.5$).

(a) Survival probability $S_n = \mathbb{P}(Z_n > 0)$, showing the late generation asymptotic scaling $S_n \sim 2/n$.

(b) Probability mass function of the extinction time $T_{\text{extinct}} = \inf\{n : Z_n = 0\}$, exhibiting a heavy tail $\mathbb{P}(T_{\text{extinct}} = n) \sim 2/n^2$.

(c) Probability mass function of the total progeny $\mathcal{T} = \sum_{k \geq 0} Z_k$, displaying the classical critical branching asymptotic $\mathbb{P}(\mathcal{T} = n) \sim n^{-3/2}$. (10^6 realizations).

For such power law distributed variables there is a modestly high chance of extraordinarily large realisations. When you compute a sample mean of a power law distributed variable, the more samples you compute, the larger the mean. It never converges because the occurrence of exceedingly large values pushes the normalised sum higher and higher.

But aside from this intuitive understanding, we will now take an aside to see how this curious result of infinite expected lifetime can be demonstrated.

1.1 Certainty of extinction and expected lifetime for critical GW branching

We will first derive that extinction is certain in the critical case, then we will go on to demonstrate that the expected lifetime of such a process is infinite.

1.1.1 Certainty of extinction

We will see that u_n , the probability of extinction by generation n , can be computed using the probability generating function. And it will follow that the probability of ultimate extinction will be 1 in both critical and sub-critical branching. $S_n = 1 - u_n$ will denote the survival probability, the likelihood of non-extinction at generation n . A computation using S_n will later demonstrate that the expected lifetime, $\mathbb{E}(T)$, is infinite in the critical case.

The probability generating function is defined by

$$\phi(s) = \mathbb{E}(s^X),$$

for a discrete random variable X .

In our case, the offspring distribution is given by $X = Z_1$, and hence

$$\phi(s) = \mathbb{E}(s^{Z_1}) = ps^2 + (1 - p).$$

For the Galton–Watson processes, it is well known that

$$u_{n+1} = \phi(u_n),$$

where u_n denotes the probability of extinction by generation n .

This relationship can be understood intuitively by considering the fate of the population after the first generation.

- If the process dies in its first step, then extinction has already occurred. In this case, the probability of extinction by any generation is equal to 1. This event occurs with probability $1 - p$.
- If the process replicates in its first step (w/ prob p), then the population reaches generation 1 with two individuals. Extinction by generation $n + 1$ then requires both of these fresh lineages to become extinct by their n th generation; and this happens w/ prob u_n^2 .

Combining these cases yields the recursion

$$u_{n+1} = (1 - p) + p u_n^2 = \phi(u_n).$$

which describes the probability of extinction from one generation to the next. And given the population cannot come back to life once it has died out, u_n will be a non-decreasing sequence bounded above by 1. Conversely, the probability of survival defined by the complement,

$$S_n = \mathbb{P}(Z_n > 0) = 1 - u_n,$$

will be monotonically decreasing and bounded below by 0, but may converge to a positive value $S_\infty \in [0, 1]$. And to determine how these probabilities behave in the late time limit, we can use the fact that they converge and seek steady state solutions.

Substituting $u_n = 1 - S_n$ into equation (1), we obtain

$$1 - S_{n+1} = p(1 - S_n)^2 + (1 - p).$$

Expanding and simplifying gives

$$S_{n+1} = 2pS_n - pS_n^2.$$

To find steady-state solutions, let

$$S_{n+1} = S_n = S_\infty.$$

Then S_∞ satisfies

$$S_\infty = 2pS_\infty - pS_\infty^2,$$

or equivalently,

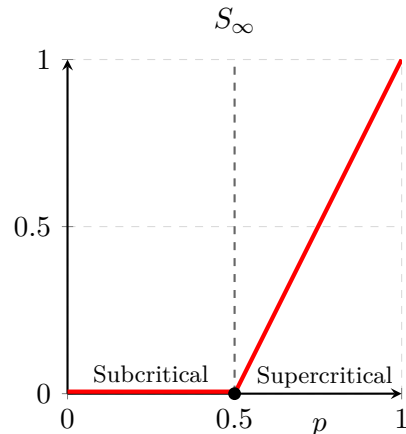
$$S_\infty(S_\infty + 1 - 2p) = 0.$$

Thus, there are two roots for S_∞ corresponding to two potential steady state solutions.

$$S_\infty = 0 \quad \text{and} \quad S_\infty = 2p - 1$$

But since S_n is a probability, it must lie within the interval $[0, 1]$, so when $p < 0.5$, the second root is negative and thus not physically meaningful. Then, when $p > 0.5$, the probability of survival is positive, given by the larger root, $S_\infty = 2p - 1$.

So we have determined that in both the subcritical ($p < 0.5$) and critical ($p = 0.5$) cases, extinction is certain. What about the expected lifetime of the process. How could it be infinite if extinction is certain?



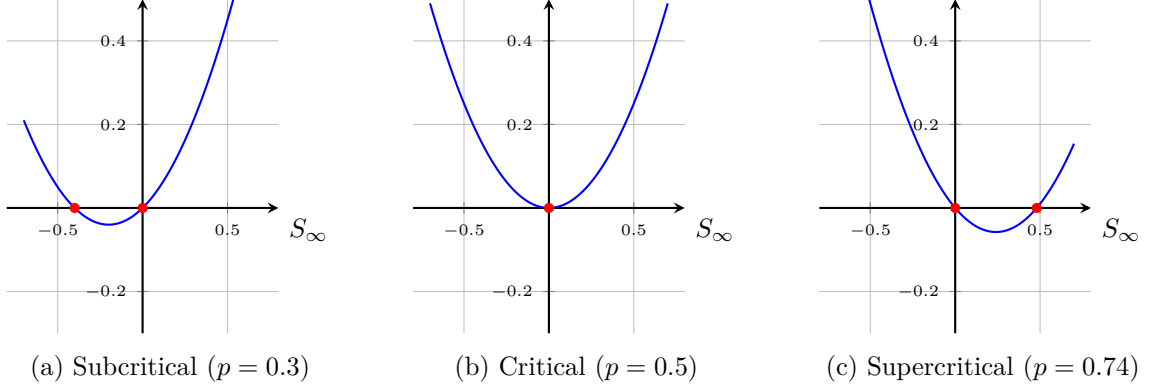


Figure 2: Visualizing $S_\infty(S_\infty + 1 - 2p) = 0$. The roots (red dots) represent equilibrium points. At $p = 0.5$, the roots collide at the origin, and for $p > 0.5$, a stable positive solution $S_\infty = 2p - 1$ emerges.

1.1.2 Expected lifetime

We are interested in $\mathbb{E}(T)$ when $p = 0.5$, where T denotes the *extinction time* or the number of generations that the Galton Watson process survives. This is a measure of how many generations the process survives and is defined as

$$T = \inf\{n : Z_n = 0\}.$$

that is, the smallest generation index n at which the population is no longer positive. The process continues over generations

$$0, 1, 2, \dots, T - 1,$$

then fixates to 0, surviving for T generations in total.

To compute the expected time to extinction, we can entertain the following representation.

$$T = \sum_{n=0}^{\infty} \mathbf{1}_{\{T > n\}},$$

where $\mathbf{1}_{\{\text{boolean}\}}$ is the indicator function, evaluated:

$$\begin{aligned} \mathbf{1}_{\{\text{true}\}} &= 1 \\ \mathbf{1}_{\{\text{false}\}} &= 0 \end{aligned}$$

The sum involving the indicator function $\mathbf{1}_{\{T > n\}}$ counts one unit for every generation in which the process is alive.

For example, if $T = 5$, then

$$\mathbf{1}_{\{T > n\}} = 1 + 1 + 1 + 1 + 1 + 0 + 0 + \dots$$

Then, to compute the expected value of T ,

$$\mathbb{E}(T) = \mathbb{E}\left(\sum_{n=0}^{\infty} \mathbf{1}_{\{T > n\}}\right) = \sum_{n=0}^{\infty} \mathbb{E}(\mathbf{1}_{\{T > n\}}),$$

And given

$$\mathbf{1}_{\{T > n\}} = \begin{cases} 1, & \text{if } T > n, \quad (\text{alive at } n) \\ 0, & \text{otherwise,} \end{cases}$$

with

$$\mathbb{E}(\mathbf{1}_{\{T > n\}}) = \mathbb{P}(T > n).$$

$T > n$ is equivalent to $Z_n > 0$; the process is still alive at generation n . Therefore,

$$\mathbb{E}(T) = \sum_{n=0}^{\infty} \mathbb{P}(Z_n > 0).$$

In other words, with $S_n = \mathbb{P}(Z_n > 0)$ being the survival probability at generation n ,

$$\mathbb{E}(T) = \sum_{n=0}^{\infty} S_n.$$

Here lies an apparent paradox. We have already established that extinction is certain in the critical case, so $S_n \rightarrow 0$ as $n \rightarrow \infty$. Yet the expected time to extinction is infinite, and $\mathbb{E}(T)$ is the sum of S_n which collapses to 0.

Explaining this will require the curious fact that the harmonic series sum is non-convergent.

$$\sum_{n=1}^{\infty} \frac{1}{n} = \infty$$

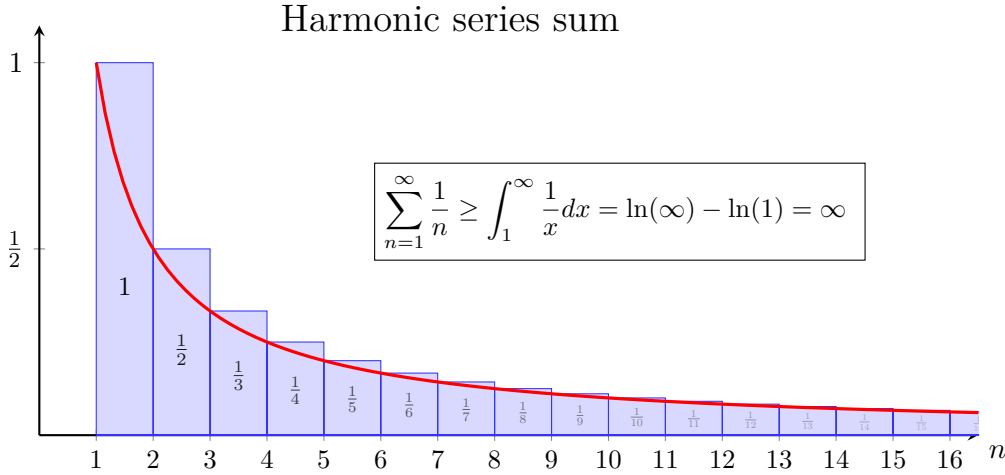


Figure 3: The harmonic series sum diverges. The sum of the areas of the blue rectangles (left) exceeds the area under the red curve (right), which diverges as $x \rightarrow \infty$. Thus, $\sum_{n=1}^{\infty} \frac{1}{n} = \infty$. This fact will be used to show that the expected time to extinction in the critical galton watson process is infinite.

To see how, we first turn our attention to the iterative formula for S_n :

$$S_{n+1} = 2pS_n - pS_n^2.$$

In the critical case, $p = 0.5$, this becomes

$$S_{n+1} = S_n - \frac{1}{2}S_n^2.$$

Rearranging gives

$$S_{n+1} - S_n = -\frac{1}{2}S_n^2.$$

Then, if we consider the late generation limit, $n \gg 1$, the unit 1 can be considered as a diminishingly small increment compared to n , and because of this, the above equation tends to meet the definition of a derivative:

$$\frac{S_{n+1} - S_n}{1} \rightarrow \frac{dS(n)}{dn} = -\frac{1}{2}S(n)^2.$$

So as $n \rightarrow \infty$, the evolution of S_n is increasingly well approximated by this ODE, and thus by its general solution,

$$S(n) = \frac{2}{n + \text{const.}}$$

Hence,

$$S_n \sim \frac{2}{n} \quad \text{as } n \rightarrow \infty, \tag{1}$$

which is supported by the numerical simulation results displayed in Figure 1(a).

Then, coming back to the question of expected lifetime in the critical case, we find that the sum tends towards 2 times the harmonic series in the late generation limit:

$$\mathbb{E}(T) = \sum_{n=0}^{\infty} S_n \sim \sum_{n=1}^{\infty} \frac{2}{n}.$$

And given S_n is always positive and that the harmonic series sum diverges, it can be concluded that $E(T)$ diverges too. Thus, the expected lifetime of the critical Galton-Watson process is infinite, despite the certainty of extinction.

2 Limiting Conditional Mean $\mathbb{E}(Z_n | Z_n > 0) = 1/A(p)$

With $0 \leq p \leq \frac{1}{2}$ (and non-degenerate offspring law), the Galton-Watson process becomes extinct almost surely. For the constant $A(p)$ we restrict to the open subcritical interval $p \in [0, \frac{1}{2})$, where $A(p) \in (0, \frac{1}{2}]$. But picture this: You initiate a large number of such branching chains with equal $p \leq 0.5$, and watch as they simultaneously progress. Many will die out early, but some may persist for many generations. Here, we will consider the behaviour of those realisations which outlive the majority.

Specifically, we will aim to compute $\mathbb{E}(Z_n | Z_n > 0)$ for $n \rightarrow \infty$, the conditional limiting mean of non-extinct populations, and we will see that it becomes stationary.

To begin, we can decompose the expectation of population size into the two conditional expectations of persistence and cessation:

$$\mathbb{E}(Z_n) = \mathbb{E}(Z_n | Z_n > 0) \cdot \mathbb{P}(Z_n > 0) + \mathbb{E}(Z_n | Z_n = 0) \cdot \mathbb{P}(Z_n = 0).$$

Since $\mathbb{E}(Z_n | Z_n = 0) = 0$, and $\mathbb{P}(Z_n > 0) = S_n$, this simplifies to

$$\mathbb{E}(Z_n) = \mathbb{E}(Z_n | Z_n > 0) \cdot S_n.$$

Hence,

$$\mathbb{E}(Z_n | Z_n > 0) = \frac{\mathbb{E}(Z_n)}{S_n}.$$

Then, we know that $\mathbb{E}(Z_n) = (2p)^n$, so

$$\mathbb{E}(Z_n | Z_n > 0) = \frac{(2p)^n}{S_n}.$$

And we understand what happens to $(2p)^n$ for large n . But what about S_n ? It may go to 0 in the limit, but how does it get there, and what will this say about the conditional mean?

Recall the recursion

$$S_{n+1} = 2p S_n - p S_n^2.$$

Because $S_n \rightarrow 0$, for growing generation count the quadratic term $p S_n^2$ diminishes faster than the linear term. Consequently, at large n the recursion is increasingly well approximated by

$$S_{n+1} \sim 2p S_n.$$

And thus there exists a constant $A(p) \in (0, \frac{1}{2}]$, depending on p but not on n , such that

$$S_n \sim A(p) (2p)^n \quad \text{as } n \rightarrow \infty, \quad (2)$$

or equivalently

$$A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{(2p)^n}. \quad (3)$$

(The limit of S_n itself is 0 for $p < \frac{1}{2}$; one must not write $\lim S_n = A(p)(2p)^n$ as an ordinary limit identity.) The constant $A(p)$ absorbs the cumulative effect of the quadratic terms in the survival recurrence before the linear regime dominates. And it follows that the conditional mean in the late generation limit is given by

$$\lim_{n \rightarrow \infty} \mathbb{E}(Z_n | Z_n > 0) = \lim_{n \rightarrow \infty} \frac{(2p)^n}{S_n} = \frac{1}{A(p)}.$$

So because $A(p)$ is constant, the limiting conditional mean is also constant; it may also be referred to as the quasi-stationary mean population.

But the constant $A(p)$ is still something of a mystery. As we will see, computing an analytic formula is not straightforward.

3 Attempting to compute $A(p)$

We can compute approximate values of $A(p)$ numerically by iterating $S_{n+1} = f(S_n)$ many times to find very near stationary values of S_n at $n \gg 1$. Then, computing $S_{n_{\text{large}}} / (2p)^{n_{\text{large}}}$ gives a value approximating $A(p)$ for that p . Repeating this over a range of probabilities can plot a curve of $A(p)$ for $p \in [0, 0.5)$. And this works reasonably well most of the time. But not when p is from very close to $p = 0.5$, because as $p \rightarrow 0.5$, the required n for S_n to reach near stationarity grows impractically large.

No elementary closed form for $A(p)$ is known. During the project, a number of approaches were tried, for example:

- Continuum approximations of the recurrence $S_{n+1} = f(S_n)$ by an ODE in generation number;
- Symbolic regression / genetic search for a closed expression fitted to numerical samples of $A(p)$.

None produced a satisfactory elementary formula. Sections below record partial analytic gains (product representation, near-critical asymptotics). The link to the Koenigs function of the logistic map, and a careful account of the obstruction to an elementary closed form, are given in Sections 4–5. Exceptional elementary linearisations exist at logistic parameters $r = 2$ and $r = 4$, but those multipliers lie *outside* the subcritical branching window $r = 2p \in [0, 1)$; see Appendix A.

3.1 Derivation of $A(p)$ as an infinite product

We may not have found a closed formula for $A(p)$, but here is a derivation of $A(p)$ represented using an infinite product.

Taking, the nonlinear map for S_k ,

$$S_{n+1} = 2pS_n - pS_n^2.$$

To begin, substitute $S_n = 2w_n$. Then

$$2w_{n+1} = 2p \cdot (2w_n) - p \cdot (2w_n)^2 = 4p w_n - 4p w_n^2,$$

so with $r = 2p$,

$$w_{n+1} = r w_n (1 - w_n) = f_r(w_n). \quad (4)$$

This is the logistic map on the unit interval. For $r \in [0, 4]$ it exhibits the familiar period-doubling cascade; subcritical branching uses only the window $r = 2p \in [0, 1)$.

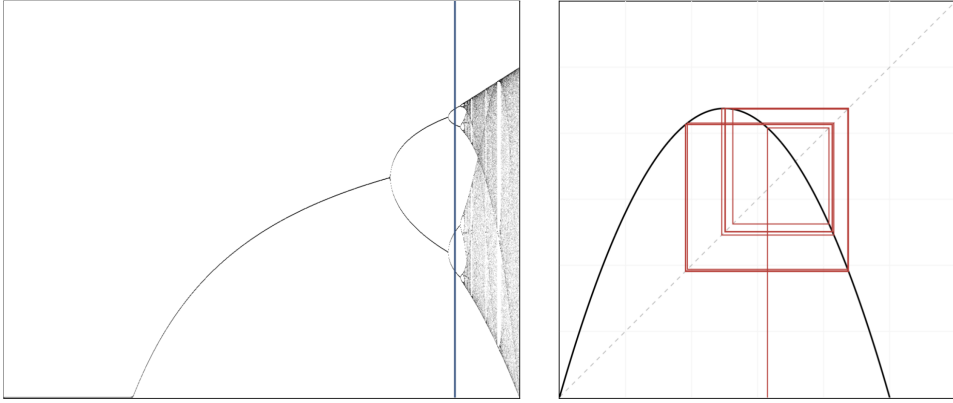


Figure 4: Phase diagram of the logistic map that shows the period doubling of stable states, ultimately resulting in chaotic behaviour. Blue sample line on the left at $r = 3.5255$, right plot shows a trajectory of the map with the same value of r .

Now, recall the form of $A(p)$,

$$A(p) = \lim_{n \rightarrow \infty} \left(\frac{S_n}{(2p)^n} \right) = 2 \lim_{n \rightarrow \infty} \left(\frac{w_n}{r^n} \right).$$

Then, because

$$x_n = x_0 \left(\frac{x_1}{x_0} \right) \left(\frac{x_2}{x_1} \right) \cdots \left(\frac{x_{n-1}}{x_{n-2}} \right) \left(\frac{x_n}{x_{n-1}} \right) = x_0 \prod_{k=0}^{n-1} \frac{x_{k+1}}{x_k},$$

we can represent an infinite limit as an infinite product:

$$\lim_{n \rightarrow \infty} (x_n) = x_0 \prod_{k=0}^{\infty} \left(\frac{x_{k+1}}{x_k} \right).$$

With $x_n = w_n/r^n$,

$$A(p) = 2 \lim_{n \rightarrow \infty} (x_n).$$

And converting this to an infinite product as above, we can simplify the fraction,

$$\frac{x_{k+1}}{x_k} = \frac{\frac{w_{k+1}}{r^{k+1}}}{\frac{w_k}{r^k}} = \frac{w_{k+1}}{w_k} \cdot \frac{r^k}{r^{k+1}}$$

$$\begin{aligned}
&= \frac{rw_k(1-w_k)}{w_k} \cdot \frac{1}{r} \\
&= 1 - w_k
\end{aligned}$$

leading to

$$A(p) = 2 \frac{w_0}{r^0} \prod_{k=0}^{\infty} (1 - w_k)$$

Then, with $w_0 = S_0/2 = 1/2$, and $r^0 = 1$, and pulling out the first ($k = 0$) factor,

$$A(p) = \frac{1}{2} \prod_{k=1}^{\infty} (1 - w_k).$$

And finally, reversing the substitution of $S_n = 2w_n$,

$$A(p) = \frac{1}{2} \prod_{k=1}^{\infty} \left(1 - \frac{S_k}{2}\right). \quad (5)$$

So here we have a different formula for $A(p)$ which can be used to progressively compute an approximation by multiplying successive factors. But this still requires producing the values of S_n by iterating the nonlinear map, so it's not clear that this result brings any material advantage.

3.1.1 Convergence of the product

For $0 < w_k < 1$, the infinite product $\prod_{k=1}^{\infty} (1 - w_k)$ converges to a strictly positive limit if and only if $\sum_{k=1}^{\infty} w_k < \infty$. From the recurrence $w_{n+1} = rw_n(1 - w_n)$ with $0 < r < 1$ and $0 < w_n < 1$,

$$w_{n+1} < rw_n.$$

By induction $w_n \leq w_0 r^n$. Since $0 < r < 1$, the bound is summable, so $\sum_k w_k < \infty$ and therefore

$$\prod_{k=1}^{\infty} (1 - w_k)$$

converges to a positive finite value. Combined with the exact product identity, this confirms that the limit defining $A(p)$ is positive and finite for each $p \in (0, \frac{1}{2})$.

3.1.2 Advantages of the infinite product formula.

3.2 Asymptotic formula for $A(p)$ as $p \rightarrow \frac{1}{2}^-$

Recall

$$S_{n+1} = 2p S_n - p S_n^2, \quad S_0 = 1,$$

and

$$\mathbb{E}(Z_n \mid Z_n > 0) = \frac{(2p)^n}{S_n} \rightarrow \frac{1}{A(p)}$$

whenever the limit defining $A(p)$ exists. Direct numerical evaluation of $S_n/(2p)^n$ becomes expensive as $p \rightarrow \frac{1}{2}^-$, because $A(p) \rightarrow 0$, the conditional mean diverges, and S_n decays only very slowly. A continuum approximation restores the leading asymptotic.

Exact difference and continuum limit. Write $\varepsilon = 1 - 2p \rightarrow 0^+$, so $p = \frac{1}{2}(1 - \varepsilon)$ and $2p = 1 - \varepsilon$. Then

$$S_{n+1} - S_n = (2p - 1)S_n - pS_n^2 = -\varepsilon S_n - pS_n^2 = -\varepsilon S_n - \frac{1}{2}S_n^2 + \frac{\varepsilon}{2}S_n^2. \quad (6)$$

(The linear coefficient is $2p - 1 = -\varepsilon$.) Treating generation number as continuous for large n and dropping the higher-order $O(\varepsilon S_n^2)$ term yields the Riccati ODE

$$\frac{dS}{dn} = -\varepsilon S - \frac{1}{2}S^2. \quad (7)$$

Solution via $u = 1/S$. Set $u = 1/S$. Then

$$\frac{du}{dn} = \varepsilon u + \frac{1}{2},$$

with integrating factor $e^{-\varepsilon n}$:

$$\frac{d}{dn}(ue^{-\varepsilon n}) = \frac{1}{2}e^{-\varepsilon n}.$$

Hence

$$u(n) = C e^{\varepsilon n} - \frac{1}{2\varepsilon}, \quad S(n) = \frac{1}{C e^{\varepsilon n} - \frac{1}{2\varepsilon}} = \frac{2\varepsilon}{2\varepsilon C e^{\varepsilon n} - 1},$$

for a constant $C > 0$ fixed by matching (not by $S_0 = 1$, since the continuum approximation is only for large n).

Matching to $S_n \sim A(p)(2p)^n$. For small ε , $(2p)^n = (1 - \varepsilon)^n \sim e^{-\varepsilon n}$. Late-time survival therefore behaves as

$$S_n \sim A(p) e^{-\varepsilon n}.$$

From the continuum solution, as $n \rightarrow \infty$,

$$S(n) \sim \frac{1}{C} e^{-\varepsilon n},$$

so $A(p) = 1/C$. Matching to the critical decay $S_n \sim 2/n$ in the double limit $\varepsilon \rightarrow 0$ with $n\varepsilon$ of order one fixes $C \sim 1/(2\varepsilon)$. Consequently

$$A(p) \sim 2\varepsilon = 2(1 - 2p) = 2 - 4p \quad \text{as } p \rightarrow \frac{1}{2}^-. \quad (8)$$

Equivalently,

$$\frac{1}{A(p)} \sim \frac{1}{2(1 - 2p)}.$$

This agrees at leading order with the continuous-time reciprocal conditional mean under the rate matching $p = \beta/(\beta + \mu)$, namely $(1 - p)/(1 - 2p) \sim 1/(2(1 - 2p))$. Formula (8) is the practical surrogate for $A(p)$ in a thin window below criticality where raw iteration of S_n is costly.

3.3 Genetic algorithm search attempts

Before the Koenigs connection was appreciated, a number of methods were tried in search of an elementary formula. And one of those was a Genetic algorithm search based regression. The concept is simple: create 50 functions at random, measure how close each one of them is to correctly describing the curve, then select the best 5 and produce a new population of 50 by

creating variations on the best 5. Measure the new 50, and repeat this cycle of variation and selection many times over.

When implementing the first version, I used a search space of combined polynomials; partly because I had a feeling that the solution might show up in that form, and partly because of the relative simplicity of encoding such functions.

In the minimal case, a basic polynomial of the standard form can be used.

$$a_0 + a_1p + a_2p^2 + a_3p^3 + a_4p^4 + \dots$$

And the "genome" encoding this class of functions would be $[a_0, \dots, a_k]$, to some arbitrarily chosen max order k . The initial 50 genomes get produced as lists of uniformly random generated real coefficients. It turns out that many different particular and substantially varying formulas will produce a curve that looks very close to correct, but we wanted the one exact formula... It occurred to me that the true formula may contain only integer or rational coefficients. So an updated version pursued polynomials of the form,

$$\frac{a_0}{b_0} + \frac{a_1}{b_1}p + \frac{a_2}{b_2}p^2 + \dots + \frac{a_k}{b_k}p^k$$

with $[a_0, \dots, a_k] \in \mathbb{Z}^k$ & $[b_0, \dots, b_k] \in \mathbb{Z}^k$. Then, after some experiments yielding no definite solution, I also tried extending the formula space to

$$\frac{\frac{a_0}{b_0} + \frac{a_1}{b_1}p + \frac{a_2}{b_2}p^2 + \dots + \frac{a_k}{b_k}p^k}{\frac{c_0}{d_0} + \frac{c_1}{d_1}p + \frac{c_2}{d_2}p^2 + \dots + \frac{c_k}{d_k}p^k}$$

with genome $[a_0, \dots, a_k, b_0, \dots, b_k, c_0, \dots, c_k, d_0, \dots, d_k] \in \mathbb{Z}^{4k}$. But after numerous searches, the best that was found merely fit the high precision numerically generated data points quite well. And the formula didn't appear simple enough to represent a probably closed solution:

$$\hat{A}_1(p) = \frac{\frac{1}{2}(1-2p)(1+2p)}{1 + \frac{1}{2}p - \frac{5}{2}p^2 - \frac{6}{5}p^3 + \frac{6}{5}p^4}$$

It started to become increasingly apparent that a huge number of very different formulas can behave almost identically in a given range.

$\hat{A}_1(p)$ does pass through (0, 0.5) & (0.5, 0) and form a pretty good approximation of the curve visually. Though it does not pass the horizontal axis with the correct gradient of -4 that we know from the asymptotic analysis above. Instead, it passes (0.5, 0) with a gradient of -3.636.

Two years after setting this work aside, I decided to have another go at using an evolutionary regression method like this one to find $A(p)$. It was just starting to become apparent that no formula exists, but I had another go anyway. This time with the option of using AI tools to write a more sophisticated symbolic tree-search program with leveraged time and effort. This new version is capable of searching a space of progressively nested elementary functions of all kinds. Although, of course, a similar conclusion was reached: many functions of varying form will closely resemble the target curve, but no satisfactory simple formula emerged, even on favouring solutions of less component parts.

Below are a couple of generated curves selected arbitrarily from the sea of similarly performing candidates.

$$\begin{aligned}\hat{A}_2(p) = & 0.31696448624690005 \ln \left(\left| \cos \left(\sqrt{\left| \sqrt{\frac{p/p}{|3.19801133824532|}} \right|} \right) \right. \right. \\ & \left. \left. + \left(\sin \left(\cos(|p|(0.621865696665606 - p) \cos(e^{e^p})) \right) + e^{\ln(|p|)} \right) p \right) \right| \right) + 0.5982282661889831\end{aligned}$$

As you can see, there are some redundancies in the formula above such as p/p . These are introduced by the program sort of like neutral mutations in biological evolution having no effect on fitness. Below is the same formula but with these removed.

$$\begin{aligned}\hat{A}_2(p) = & 0.31696448624690005 \ln \left(\left| \cos \left(\left(\frac{1}{3.19801133824532} \right)^{1/4} \right. \right. \right. \\ & \left. \left. + p \left[\sin \left(\cos(|p|(0.621865696665606 - p) \cos(e^{e^p})) \right) + |p| \right] \right) \right| \right) + 0.5982282661889831\end{aligned}$$

The next one is a bit more concise:

$$\hat{A}_3(p) = \frac{-0.616647881739505 e^{-0.552605335919693 e^p}}{\sin(\cos p) - |p|} + 0.921964323679771$$

PLOTIT

But a big problem with this formula and others produced by this program is that they either do not pass through the horizontal axis at $(0.5, 0)$ or they do not have the correct gradient of -4 at the critical point. And these discrepancies cause big inaccuracies when computing the quasistationary mean with $1/A$ as p approaches 0.5 and A draws ever closer to 0 . So whilst making the program which plots many Galton-Watson trajectories along with the horizontal red dashed line indicating the QS mean, I played around with a combined approach seen in $\hat{A}_4(p)$ which aims to correct the near critical behaviour with the asymptotic formula.

$$\hat{A}_4(p) = \begin{cases} -0.04011483983619545 e^{e^{2p}} + 0.6079994336528085, & p < 0.499962, \\ 2 - 4p, & p \geq 0.499962. \end{cases}$$

This essentially worked, but I found a better solution in a combination of the numerically generated values in combination with the asymptotic $2 - 4p$ in a narrow region of p close to 0.5 .

4 Relationship to the Koenigs function

We now make precise the link between $A(p)$ and the classical linearisation of the logistic map, and then import the content of the “no closed form” analysis for the conjugacy on $r \in (0, 1)$.

4.1 From the survival recurrence to the logistic map

Recall

$$S_{n+1} = 2p S_n - p S_n^2, \quad S_0 = 1,$$

and $A(p) = \lim_{n \rightarrow \infty} S_n / (2p)^n$. Setting $S_n = 2w_n$ and $r = 2p$ yields the logistic iteration

$$w_{n+1} = f_r(w_n), \quad f_r(w) = rw(1 - w), \quad w_0 = \frac{1}{2}, \quad (9)$$

with $r \in [0, 1)$ throughout the subcritical regime. Then

$$A(p) = 2 \lim_{n \rightarrow \infty} \frac{w_n}{r^n}. \quad (10)$$

4.2 Koenigs linearisation

Definition 4.1 (Koenigs function for the logistic map). Let $f_r(z) = rz(1 - z)$ with multiplier r at the fixed point 0 satisfying $0 < |r| < 1$. There exists a unique holomorphic map ψ_r , defined in a neighbourhood of 0, such that

$$\psi_r(f_r(z)) = r \psi_r(z), \quad \psi_r(0) = 0, \quad \psi_r'(0) = 1. \quad (11)$$

Equivalently,

$$\psi_r(z) = \lim_{n \rightarrow \infty} r^{-n} f_r^{\circ n}(z) \quad (12)$$

uniformly near the origin [1]. The normalisation $\psi_r'(0) = 1$ is equivalent to $\psi_r(z)/z \rightarrow 1$ as $z \rightarrow 0$.

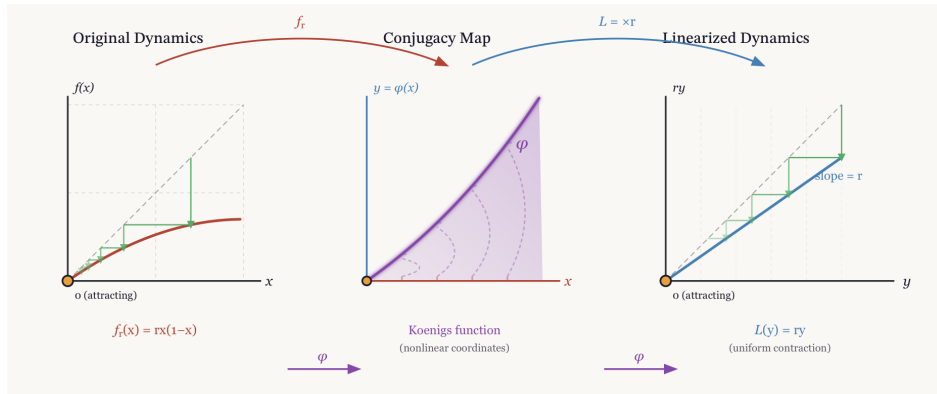


Figure 5: Schematic of Koenigs linearisation: f_r is conjugate, via ψ_r , to multiplication by r near the origin.

4.3 The identity $A(p) = 2 \lim_{n \rightarrow \infty} r^{-n} f_r^{\circ n}(\frac{1}{2})$

The local Koenigs function is initially defined only in a neighbourhood of 0. For $r \in (0, 1)$ the origin is attracting on the real line and the interval $(0, 1)$ lies in its real basin: starting from $w_0 = \frac{1}{2}$, one has $w_n \rightarrow 0$. On that basin the limit

$$\Psi_r(w_0) := \lim_{n \rightarrow \infty} r^{-n} f_r^{\circ n}(w_0) \quad (13)$$

exists and agrees with the local conjugacy near 0 (where $\psi_r'(0) = 1$ means $\psi_r(w)/w \rightarrow 1$ as $w \rightarrow 0$, so $\psi_r(w_n)/r^n \sim w_n/r^n$). Along the orbit, whenever w_n lies in the local domain,

$$\psi_r(w_0) = \frac{\psi_r(w_n)}{r^n};$$

passing to the limit yields $\Psi_r(w_0) = \lim w_n/r^n$. With $w_0 = \frac{1}{2}$ and (10),

$$A(p) = 2\Psi_r(\frac{1}{2}) = 2 \lim_{n \rightarrow \infty} r^{-n} f_r^{\circ n}(\frac{1}{2}), \quad r = 2p \in [0, 1). \quad (14)$$

We write $A(p) = 2\psi_r(\frac{1}{2})$ as a shorthand for this basin value of the Koenigs conjugacy, not as a naive evaluation of a power series beyond its guaranteed disc of convergence.

Thus every question about an elementary closed form for $A(p)$ is tied to the nature of that conjugacy at multiplier $r = 2p$.

5 No elementary closed form for the conjugacy when $r \in (0, 1)$

The analysis of this section follows the note on hypertranscendence of the logistic Koenigs function [1, 2], integrated here as the natural continuation of the $A(p)$ story.

5.1 Statement

For $r \in (0, 1)$ the origin is an attracting fixed point of f_r with multiplier $\lambda = r$. Classical linearisation supplies ψ_r as in Definition 4.1. Results of Becker and Bergweiler on hypertranscendence of conjugacies for rational maps [2] imply that such linearisations are differentially algebraic only for highly exceptional normal forms (monomial or Chebyshev type). Under the standard affine conjugacy of f_r to $z^2 + c$ with

$$c(r) = \frac{2r - r^2}{4}, \quad (15)$$

the interval $r \in (0, 1)$ maps to $c \in (0, \frac{1}{4})$, which avoids the exceptional quadratic parameters $c \in \{0, -2\}$ associated with $r = 2$ and $r = 4$. Therefore, for each fixed $r \in (0, 1)$, the conjugacy ψ_r is hypertranscendental as a function of z : it satisfies no algebraic differential equation and is not elementary in the Liouville sense.

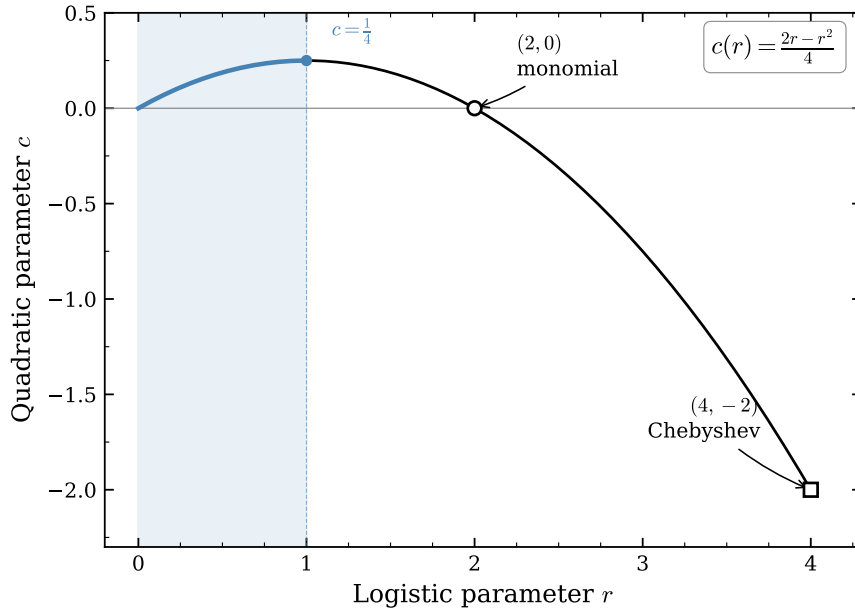


Figure 6: Parameter correspondence $c(r) = (2r - r^2)/4$. The segment $r \in (0, 1)$ maps to $c \in (0, \frac{1}{4})$, disjoint from the integrable values $c \in \{0, -2\}$.

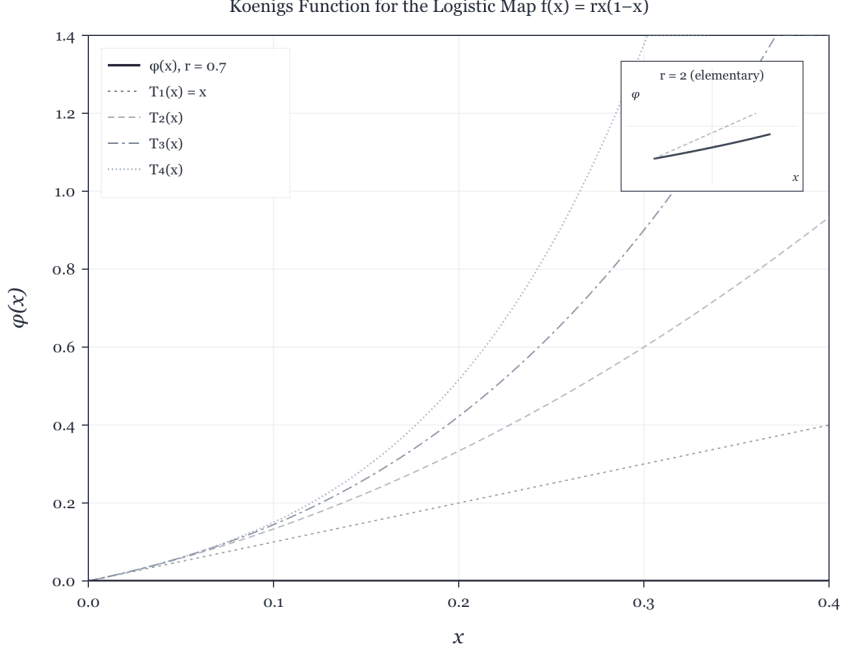


Figure 7: Numerical approximation of a Koenigs function for a logistic multiplier in $(0, 1)$, with low-order Taylor polynomials near the origin. Elementary closed forms exist only for exceptional multipliers outside this range (Appendix A).

5.2 Consequences for $A(p)$

Combined with (14), extracting $A(p)$ cannot proceed by evaluating a single elementary expression for ψ_{2p} as a function of z . Two logical caveats remain if one wants a stronger theorem:

1. Non-elementarity of $z \mapsto \psi_r(z)$ does not by itself forbid a closed form for one value $\psi_r(\frac{1}{2})$ (compare Γ and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$).
2. Elevating the statement to a full classification of the scalar map $p \mapsto A(p)$ is a further step beyond fixed- r hypertranscendence.

The working claim of this note is therefore:

No elementary closed-form expression for $A(p)$ on $p \in [0, \frac{1}{2})$ is known, and none is expected: $A(p)$ is tied to the linearisation of the logistic map at multiplier $r = 2p \in [0, 1)$, in a regime where that conjugacy is non-elementary for each fixed $r \in (0, 1)$, while the classical elementary logistic cases $r = 2, 4$ lie outside the subcritical branching range.

Full derivations of the elementary solutions at $r = 2$ and $r = 4$, and a more detailed account of the conjugacy and the Becker–Bergweiler dichotomy, are collected in Appendix A.

6 Practical computation of $A(p)$ and $1/A(p)$

In the absence of an elementary formula, $A(p)$ is obtained by iteration of $S_{n+1} = 2pS_n - pS_n^2$ (or the product (5)) for p bounded away from $\frac{1}{2}$, and by the near-critical asymptotic $A(p) \approx 2 - 4p$

in a narrow window as $p \rightarrow \frac{1}{2}^-$. A robust hybrid scheme is

$$A(p)_{\text{prox}} = \begin{cases} \frac{1}{2} \prod_{k=1}^N \left(1 - \frac{S_k}{2}\right), & p < c, \quad N \gg 1, \\ 2 - 4p, & c \leq p < \frac{1}{2}, \end{cases}$$

with a cutoff c slightly below $\frac{1}{2}$ (e.g. 0.4999). This restores accurate quasi-stationary means $1/A(p)$ even where raw iteration becomes expensive.

7 Conclusion

The constant $A(p)$ organises the quasi-stationary mean of subcritical binary Galton–Watson branching. It is well defined, computable, and linked to the basin Koenigs value $A(p) = 2 \lim r^{-n} f_r^{\circ n}(\frac{1}{2})$ at multiplier $r = 2p$. The hypertranscendence theory for that conjugacy on $r \in (0, 1)$ explains the failure of elementary closed-form searches, while leaving $A(p)$ fully usable for analysis and numerics. Exceptional elementary linearisations at $r = 2$ and $r = 4$ exist but lie outside the branching window; their derivations appear in the appendix.

A Elementary Koenigs functions at $r = 2$ and $r = 4$

These cases lie outside subcritical branching ($r = 2p < 1$) but complete the classical picture of integrable logistic linearisations. The derivations follow the closed-form note integrated into this document.

A.1 Case $r = 2$ (logarithmic)

For $r = 2$, $f_2(x) = 2x(1 - x)$. The fixed point 0 has multiplier $\lambda = 2$ (repelling); the linearisation is of Poincaré type rather than an attracting-Koenigs function, but an elementary closed form still exists.

The affine change $u = 1 - 2x$ conjugates the dynamics to $u \mapsto u^2$. Linearising $u \mapsto u^2$ by the logarithm and returning to x yields the candidate

$$\psi_2(x) = -\frac{1}{2} \ln(1 - 2x), \quad (16)$$

normalised so that $\psi_2(0) = 0$ and $\psi_2'(0) = 1$.

Verification.

$$\psi_2(f_2(x)) = -\frac{1}{2} \ln(1 - 2 \cdot 2x(1 - x)) = -\frac{1}{2} \ln((1 - 2x)^2) = -\ln(1 - 2x) = 2\psi_2(x).$$

A.2 Case $r = 4$ (inverse trigonometric)

For $r = 4$, $f_4(x) = 4x(1 - x)$. With $x = \sin^2 \theta$ one finds $f_4(x) = \sin^2(2\theta)$, so the angle doubles. The candidate

$$\psi_4(x) = (\arcsin \sqrt{x})^2 \quad (17)$$

satisfies $\psi_4(f_4(x)) = 4\psi_4(x)$ on a suitable real domain, and $\psi_4(x) \sim x$ as $x \rightarrow 0$, so $\psi_4'(0) = 1$.

Verification. Using $\arcsin(2u\sqrt{1-u^2}) = 2\arcsin u$ with $u = \sqrt{x}$ (on an appropriate interval),

$$\psi_4(f_4(x)) = (\arcsin \sqrt{4x(1-x)})^2 = (2\arcsin \sqrt{x})^2 = 4\psi_4(x).$$

A.3 Affine conjugacy to $z^2 + c$

Lemma A.1. *The logistic map $f_r(x) = rx(1 - x)$ is affinely conjugate to $P_c(z) = z^2 + c$ with $c = (2r - r^2)/4$.*

Proof sketch. Seek $h(x) = ax + b$ with $h \circ f_r \circ h^{-1} = P_c$. The substitution $x = -z/r + \frac{1}{2}$ (for $r \neq 0$) yields a monic quadratic; matching coefficients gives $c = (2r - r^2)/4$. $\square$

Under this conjugacy, $r = 2 \Leftrightarrow c = 0$ (monomial z^2) and $r = 4 \Leftrightarrow c = -2$ (Chebyshev type). Neither exceptional value lies in the image of $r \in (0, 1)$, which is $c \in (0, \frac{1}{4})$.

A.4 Becker–Bergweiler dichotomy (summary)

Theorem A.2 (Becker and Bergweiler [2]). *For a rational map of degree $d \geq 2$, the linearisation at a fixed point with multiplier λ in the classical non-resonant range is differentially algebraic only in exceptional normal forms (monomial / Chebyshev type up to analytic conjugacy).*

Applied to the quadratic family via Lemma A.1, the only elementary logistic parameters of this type are $r = 2$ and $r = 4$, both outside $r \in (0, 1)$. Hence for each fixed $r \in (0, 1)$ the conjugacy is hypertranscendental as a function of z , which underpins the main-text claim for $A(p) = 2\psi_{2p}(\frac{1}{2})$.

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